

Fig. 2: An actual-size, single-side PCB for the simple tester for 74xx04 and 74xx14 ICs

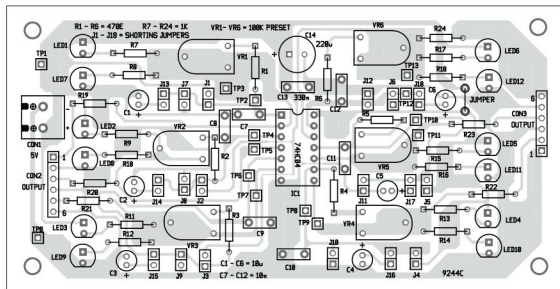


Fig. 3: Component layout for the PCB

PARTS LIST

Semiconductors:

IC1 - 74HC04 hex inverter

LED1-LED12 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R6 - 470-ohm

R7-R24 - 1-kilo-ohm

VR1-VR6 - 100-kilo-ohm preset

Capacitors:

C1-C6 - 10 μ F, 16V electrolytic

C7-C12 - 10nF ceramic disk

C13 - 330nF ceramic disk

C14 - 220 μ F, 16V electrolytic

Miscellaneous:

J1-J18

- Shorting jumper plus

- 2-pin berg strip (male)

- 14-pin IC base (socket)

CON1

- 2-pin terminal connector

CON2-CON3

- 6-pin terminal connector

high and the respective output is low. Then, LED1 through LED6 will glow. If jumpers J7 through J12 are closed, the

input of the corresponding inverter is low and the respective output is high. Then, LED7 through LED12 will glow. If any one of the LEDs does not glow, it means that the corresponding gate is damaged.

Dynamic testing. During dynamic testing, jumpers J1 through J12 should be kept open. When jumpers J13 through J18 are closed, oscillations are generated on the output of each gate. The frequency varies from 0.5Hz to 10Hz. Here, both the LEDs will blink alternately. Their blinking frequency can be changed with VR1 to VR6, if needed.

If a gate's (say, N2) output LEDs do not glow, it means that the gate's (N2's) segment is damaged.

IC 74xx04 may not oscillate during dynamic testing because it does not have specified internal Schmitt triggers. So, only IC 74xx14 can be checked dynamically. If all LEDs blink alternately, it means your 74xx14 is fine.

The circuit works on a 5V power supply but you can use a higher voltage supply, within limits specified in the datasheet.

Construction and testing

An actual-size, single-side PCB for the simple tester using a socket for the 74xx04 and 74xx14 ICs is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable plastic case.

Fix two 6-pin connectors CON2 and CON3 for six outputs. Also fix 2-pin terminal connector CON1 for power supply. Jumpers J1 through J18 are used for external shorting. You have to use eighteen 2-pin berg strip male connectors on the PCB for shorting the jumpers. Fix presets

VR1 through VR6 on the PCB for frequency control.

If you want to test an IC, insert it in the 14-pin IC socket. Connectors CON2 and CON3 are provided on the PCB for testing the voltages at each individual gate. Before inserting IC in its socket, verify the test point voltages given in the table (only for inverter gate N1 given here). Similarly, you can verify other internal gates at the respective test points shown in the circuit diagram as well as in the PCB. ●

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